Red Hat OpenShift AI Self-Managed 2.25

Upgrading OpenShift AI Self-Managed in a disconnected environment

Upgrade Red Hat OpenShift AI on OpenShift in a disconnected environment

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Abstract

Learn about the Red Hat OpenShift AI Operator upgrade process in a disconnected environment.

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PREFACE

As a cluster administrator, you can configure either automatic or manual upgrade of the OpenShift AI Operator.

CHAPTER 1. OVERVIEW OF UPGRADING OPENSHIFT AI SELF-MANAGED

As a cluster administrator, you can configure either automatic or manual upgrades for the Red Hat OpenShift AI Operator in a disconnected environment. A disconnected environment is a network restricted environment where Operator Lifecycle Manager (OLM) cannot access the default OperatorHub and image registries, which require Internet connectivity.

NOTE

For information about upgrading OpenShift AI as self-managed software on your OpenShift cluster in a connected environment, see Upgrading OpenShift AI Self-Managed.

- If you configure automatic upgrades, when a new version of the Red Hat OpenShift AI Operator is available, and you have updated your mirror registry content, Operator Lifecycle Manager (OLM) automatically upgrades the running instance of your Operator without human intervention.

- If you configure manual upgrades, when a new version of the Red Hat OpenShift AI Operator is available and you have updated your mirror registry content, OLM creates an update request. A cluster administrator must manually approve the update request to update the Operator to the new version. See Manually approving a pending Operator upgrade for more information about approving a pending Operator upgrade.

- By default, the Red Hat OpenShift AI Operator follows a sequential update process. This means that if there are several minor versions between the current version and the version that you plan to upgrade to, Operator Lifecycle Manager (OLM) upgrades the Operator to each of the minor versions before it upgrades it to the final, target version. If you configure automatic upgrades, OLM automatically upgrades the Operator to the latest available version, without human intervention. If you configure manual upgrades, a cluster administrator must manually approve each sequential update between the current version and the final, target version.
  To view information regarding the supported and tested upgrade paths for Red Hat OpenShift AI, see Red Hat OpenShift AI Upgrade Path Information.

For information about OpenShift AI Self-Managed release types and supported versions, see the Red Hat OpenShift AI Self-Managed Life Cycle Knowledgebase article.

- Before you upgrade OpenShift AI, you should complete the Requirements for upgrading OpenShift AI.

- Before you can use an accelerator in OpenShift AI, your instance must have the associated accelerator profile or hardware profile. If your OpenShift instance has an accelerator, its accelerator profile or hardware profile is preserved after an upgrade. For more information about accelerators, see Working with accelerators.

IMPORTANT

By default, hardware profiles are hidden in the dashboard navigation menu and user interface, while accelerator profiles remain visible. In addition, user interface components associated with the deprecated accelerator profiles functionality are still displayed. To show the **Settings → Hardware profiles** option in the dashboard navigation menu, and the user interface components associated with hardware profiles, set the **disableHardwareProfiles** value to **false** in the **OdhDashboardConfig** custom resource (CR) in OpenShift. For more information about setting dashboard configuration options, see Customizing the dashboard.

- Workbench images are integrated into the image stream during the upgrade and subsequently appear in the OpenShift AI dashboard.

NOTE

Workbench images are constructed externally; they are prebuilt images that undergo quarterly changes and they do not change with every OpenShift AI upgrade.

Additional resources

- Operator Lifecycle Manager workflow

CHAPTER 2. CONFIGURING THE UPGRADE STRATEGY FOR OPENSHIFT AI

As a cluster administrator, you can configure either an automatic or manual upgrade strategy for the Red Hat OpenShift AI Operator.

IMPORTANT

By default, the Red Hat OpenShift AI Operator follows a sequential update process. This means that if there are several versions between the current version and the version that you intend to upgrade to, Operator Lifecycle Manager (OLM) upgrades the Operator to each of the intermediate versions before it upgrades it to the final, target version. If you configure automatic upgrades, OLM automatically upgrades the Operator to the *latest* available version, without human intervention. If you configure manual upgrades, a cluster administrator must manually approve each sequential update between the current version and the final, target version.

For information about supported versions, see the Red Hat OpenShift AI Self-Managed Life Cycle Knowledgebase article.

Prerequisites

- You have cluster administrator privileges for your OpenShift cluster.
- The Red Hat OpenShift AI Operator is installed.
- If you previously used the oc-mirror plugin v1 to mirror images, you have migrated to the oc-mirror plugin v2 (the oc-mirror plugin v1 is now deprecated). For more information, see Migrating from oc-mirror plugin v1 to v2 in the OpenShift documentation.
- You have mirrored the required container images to a private registry. See Mirroring images to a private registry for a disconnected installation.

Procedure

1. Log in to the OpenShift cluster web console as a cluster administrator.
2. In the **Administrator** perspective, in the left menu, select **Operators → Installed Operators**.
3. Click the **Red Hat OpenShift AI** Operator.
4. Click the **Subscription** tab.
5. Under **Update approval**, click the pencil icon and select one of the following update strategies:
   - **Automatic**: New updates are installed as soon as they become available.
   - **Manual**: A cluster administrator must approve any new update before installation begins.
6. Click **Save**.

Additional resources

- For more information about the subscription channels that are available in version 2 of the Red Hat OpenShift AI Operator, see Installing the Red Hat OpenShift AI Operator.

- For more information about upgrading Operators that have been installed by using OLM, see Updating installed Operators in the OpenShift documentation.

CHAPTER 3. REQUIREMENTS FOR UPGRADING OPENSHIFT AI

When upgrading OpenShift AI in a disconnected environment, you must complete the following tasks.

Check the components in the **DataScienceCluster** object

When you upgrade Red Hat OpenShift AI, the upgrade process automatically uses the values from the previous **DataScienceCluster** object.

After the upgrade, you should inspect the **DataScienceCluster** object and optionally update the status of any components as described in Updating the installation status of Red Hat OpenShift AI components by using the web console.

NOTE

New components are not automatically added to the **DataScienceCluster** object during upgrade. If you want to use a new component, you must manually edit the **DataScienceCluster** object to add the component entry.

NOTE

If you are upgrading OpenShift AI on a cluster running in FIPS mode, any custom container images for data science pipelines must be based on UBI 9 or RHEL 9. This ensures compatibility with FIPS-approved pipeline components and prevents errors related to mismatched OpenSSL or GNU C Library (glibc) versions.

Migrate from embedded Kueue to Red Hat build of Kueue

The embedded Kueue component for managing distributed workloads is deprecated. OpenShift AI now uses the Red Hat build of Kueue Operator to provide enhanced workload scheduling for distributed training, workbench, and model serving workloads.

Before upgrading OpenShift AI, check if your environment is using the embedded Kueue component by verifying the **spec.components.kueue.managementState** field in the **DataScienceCluster** custom resource. If the field is set to **Managed**, you must complete the migration to the Red Hat build of Kueue Operator to avoid controller conflicts and ensure continued support for queue-based workloads.

> **IMPORTANT**
>
> As part of the migration to Red Hat build of Kueue, you must manually delete the following legacy Kueue CRDs:
>
> - **cohorts.kueue.x-k8s.io/v1alpha1**
> - **topologies.kueue.x-k8s.io/v1alpha1**
>
> If you have existing instances of these CRDs, you must manually back up their data, delete the instances, and recreate them using the **v1beta1** API after the upgrade. If you do not complete these steps, the Kueue Operator enters a failed reconciliation loop, resulting in a **Not Ready** status for the **DataScienceCluster**. To avoid this conflict, ensure no active workloads depend on the legacy Kueue resources.
>
> For more information, see Red Hat Build of Kueue 1.2 installation or upgrade fails with Kueue CRD reconciliation error.

This migration requires OpenShift 4.18 or later. For more information, see Migrating to the Red Hat build of Kueue Operator.

Address KServe requirements

For the KServe component, which is used by the single-model serving platform to serve large models, you must meet the following requirements:

- To fully install and use KServe, you must also install Operators for Red Hat OpenShift Serverless and Red Hat OpenShift Service Mesh and perform additional configuration. For more information, see Serving large models.
- If you want to add an authorization provider for the single-model serving platform, you must install the **Red Hat - Authorino** Operator. For more information, see Adding an authorization provider for the single-model serving platform.
- If you have *not* enabled the KServe component (that is, you set the value of the **managementState** field to **Removed** in the **DataScienceCluster** object), you must also disable the dependent Service Mesh component to avoid errors. See Disabling KServe dependencies.

Address RAG dependencies

If you plan to deploy Retrieval-Augmented Generation (RAG) workloads by using Llama Stack, you must meet the following requirements:

- You have GPU-enabled nodes available on your cluster and you have installed the Node Feature Discovery Operator and NVIDIA GPU Operator. For more information, see Installing the Node Feature Discovery Operator and Enabling NVIDIA GPUs.
- You have access to storage for your model artifacts.
- You have met the KServe installation prerequisites.

Verify Argo Workflows compatibility

If you use your own Argo Workflows instance for pipelines, verify that the installed version is compatible with this release of OpenShift AI. For details, see Supported Configurations.

Update workflows interacting with **OdhDashboardConfig** resource

Previously, cluster administrators used the **groupsConfig** option in the **OdhDashboardConfig** resource to manage the OpenShift groups (both administrators and non-administrators) that can access the OpenShift AI dashboard. Starting with OpenShift AI 2.17, this functionality has moved to the **Auth** resource. If you have workflows (such as GitOps workflows) that interact with **OdhDashboardConfig**, you must update them to reference the **Auth** resource instead.

Table 3.1. User management resource update

	OpenShift AI 2.16 and earlier	OpenShift AI 2.17 and later
apiVersion	**opendatahub.io/v1alpha**	**services.platform.opendatahub.io/v1alpha1**
kind	**OdhDashboardConfig**	**Auth**
name	**odh-dashboard-config**	**auth**
Admin groups	**spec.groupsConfig.adminGroups**	**spec.adminGroups**
User groups	**spec.groupsConfig.allowedGroups**	**spec.allowedGroups**

Check the status of certificate management

You can use self-signed certificates in OpenShift AI.

After you upgrade, check the management status for Certificate Authority (CA) bundles as described in Working with certificates.

Additional resources

- Red Hat OpenShift AI Upgrade Path Information

CHAPTER 4. UPDATING THE INSTALLATION STATUS OF RED HAT OPENSHIFT AI COMPONENTS BY USING THE WEB CONSOLE

You can use the OpenShift web console to update the installation status of components of Red Hat OpenShift AI on your OpenShift cluster.

IMPORTANT

If you upgraded OpenShift AI, the upgrade process automatically used the values of the previous version's **DataScienceCluster** object. New components are not automatically added to the **DataScienceCluster** object.

After upgrading OpenShift AI:

- Inspect the default **DataScienceCluster** object to check and optionally update the **managementState** status of the existing components.
- Add any new components to the **DataScienceCluster** object.

Prerequisites

- The Red Hat OpenShift AI Operator is installed on your OpenShift cluster.
- You have cluster administrator privileges for your OpenShift cluster.

Procedure

1. Log in to the OpenShift web console as a cluster administrator.
2. In the web console, click **Operators** → **Installed Operators** and then click the **Red Hat OpenShift AI** Operator.
3. Click the **Data Science Cluster** tab.
4. On the **DataScienceClusters** page, click the **default-dsc** object.
5. Click the **YAML** tab.
   An embedded YAML editor opens showing the default custom resource (CR) for the **DataScienceCluster** object, similar to the following example:

```
apiVersion: datasciencecluster.opendatahub.io/v1
kind: DataScienceCluster
metadata:
  name: default-dsc
spec:
  components:
    codeflare:
      managementState: Removed
    dashboard:
      managementState: Removed
    datasciencepipelines:
      managementState: Removed
    kserve:
```

```
    managementState: Removed
  kueue:
    managementState: Removed
  llamastackoperator:
    managementState: Removed
  modelmeshserving:
    managementState: Removed
  ray:
    managementState: Removed
  trainingoperator:
    managementState: Removed
  trustyai:
    managementState: Removed
  workbenches:
    managementState: Removed
    workbenchNamespace: rhods-notebooks
```

6. In the **spec.components** section of the CR, for each OpenShift AI component shown, set the value of the **managementState** field to either **Managed** or **Removed**. These values are defined as follows:

   Managed

   The Operator actively manages the component, installs it, and tries to keep it active. The Operator will upgrade the component only if it is safe to do so.

   Removed

   The Operator actively manages the component but does not install it. If the component is already installed, the Operator will try to remove it.

   IMPORTANT

   - To learn how to install the KServe component, which is used by the single-model serving platform to serve large models, see Installing the single-model serving platform.
   - If you have *not* enabled the KServe component (that is, you set the value of the **managementState** field to **Removed**), you must also disable the dependent Service Mesh component to avoid errors. See Disabling KServe dependencies.
   - To learn how to install the distributed workloads feature, see Installing the distributed workloads components.
   - To learn how to run distributed workloads in a disconnected environment, see Running distributed data science workloads in a disconnected environment.

7. Click **Save**.
   For any components that you updated, OpenShift AI initiates a rollout that affects all pods to use the updated image.

8. If you are upgrading from OpenShift AI 2.19 or earlier, upgrade the Authorino Operator to the **stable** update channel, version 1.2.1 or later.

IMPORTANT

If you are upgrading the Authorino Operator to the **stable** update channel, version 1.2.1 or later in a disconnected environment, use the following upgrade procedure described in the release notes: RHOAIENG-24786 - Upgrading the Authorino Operator from Technical Preview to Stable fails in disconnected environments. Otherwise, the upgrade can fail.

Verification

1. Confirm that there is at least one running pod for each component:
    a. In the OpenShift web console, click **Workloads** → **Pods**.
    b. In the **Project** list at the top of the page, select **redhat-ods-applications** or your custom applications namespace.
    c. In the applications namespace, confirm that there are one or more running pods for each of the OpenShift AI components that you installed.
2. Confirm the status of all installed components:
    a. In the OpenShift web console, click **Operators** → **Installed Operators**.
    b. Click the **Red Hat OpenShift AI** Operator.
    c. Click the **Data Science Cluster** tab and select the **DataScienceCluster** object called **default-dsc**.
    d. Select the **YAML** tab.
    e. In the **status.installedComponents** section, confirm that the components you installed have a status value of **true**.

    NOTE

    If a component shows with the **component-name: {}** format in the **spec.components** section of the CR, the component is not installed.

3. In the OpenShift AI dashboard, users can view the list of the installed OpenShift AI components, their corresponding source (upstream) components, and the versions of the installed components, as described in Viewing installed OpenShift AI components.